Lab 7: Fuzzing for Input Validation Bugs

Project: L6_L7_50 - Recycling Rewards | Tool: Hypothesis + pytest

Functions under test

- parse_collection("material:kg")
- weight_average("w1,w2,...")
- net_points("points|penalty")

Bugs found by fuzzing (starter code)

- None / bytes / non-string input -> AttributeError / TypeError on .split()
- Missing or extra delimiter ('', 'a:b:c') -> ValueError on tuple unpacking
- Empty or non-numeric fields (':', '|', '5,,x') -> ValueError in float()/int()
- Huge values ('1e308|0') -> net_points returns inf (overflow)
- Negative weights and nan/inf values accepted silently

Fixes applied (recycling_rewards.py)

- Type check: non-str input returns a safe default (None / 0 / 0.0)
- Exactly one delimiter required; fields stripped and must be non-empty
- Numeric conversion wrapped in try/except ValueError
- Reject non-finite (nan/inf) and negative values; guard overflow in net_points
- Valid behaviour unchanged: paper:12 -> ('Paper', 12.0), 5,12,20 -> 12, 80|10 -> 150.0

Fuzz tests (test_recycling_rewards.py)

- 3 'never crashes' properties fed text, None, ints, floats (nan/inf), bytes, lists
- 3 output-contract properties (type / non-negative / finite results)
- 3 example tests confirming valid inputs still work
- Result: starter code 7 failed / 2 passed -> fixed code 9 passed

Step 2 - Buggy application run

Running the starter recycling_rewards.py: valid input works, but malformed input ('paper') crashes with ValueError.

PowerShell - Lab 7 - starter recycling_rewards.py

```
> python recycling_rewards.py   (valid inputs)
==== Running parse_collection ====
Enter material:weight in kg (e.g. paper:12): Material and Weight: ('Paper', 12.0)

==== Running weight_average ====
Enter weights in kg as CSV (e.g. 5,12,20): Average Weight: 12

==== Running net_points ====
Enter points|penalty (e.g. 80|10): Net Points: 150.0


> python recycling_rewards.py   (malformed input: "paper")
==== Running parse_collection ====
Enter material:weight in kg (e.g. paper:12): Traceback (most recent call last):
  File "...\Lab7_starter_copy\recycling_rewards.py", line 31, in <module>
    main()
    ~~~~^^
  File "...\Lab7_starter_copy\recycling_rewards.py", line 17, in main
    result = parse_collection(user_input)
  File "...\Lab7_starter_copy\recycling_rewards.py", line 2, in parse_collection
    material, kg = text.split(':')
    ^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4 - Fuzz tests failing on buggy code

Hypothesis tests from test_recycling_rewards.py run against the starter code: 7 failed, 2 passed.

PowerShell - Lab 7 - pytest (starter code)

```
> python -m pytest -v --tb=no test_recycling_rewards.py
========================================= test session starts =========================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python313\python.exe
hypothesis profile 'default'
rootdir: ...\Lab7_starter_copy
plugins: hypothesis-6.170.15
collecting ... collected 9 items

test_recycling_rewards.py::test_parse_collection_never_crashes FAILED                    [ 11%]
test_recycling_rewards.py::test_weight_average_never_crashes FAILED                      [ 22%]
test_recycling_rewards.py::test_net_points_never_crashes FAILED                          [ 33%]
test_recycling_rewards.py::test_parse_collection_result_is_valid_or_none FAILED          [ 44%]
test_recycling_rewards.py::test_weight_average_is_nonnegative_integer FAILED             [ 55%]
test_recycling_rewards.py::test_net_points_is_finite_and_nonnegative FAILED              [ 66%]
test_recycling_rewards.py::test_parse_collection_valid_input PASSED                      [ 77%]
test_recycling_rewards.py::test_weight_average_valid_input PASSED                        [ 88%]
test_recycling_rewards.py::test_net_points_valid_and_clamped_inputs FAILED               [100%]

====================================== short test summary info ======================================
FAILED test_recycling_rewards.py::test_parse_collection_never_crashes - ExceptionGroup: Hypothesis found 3 ...
FAILED test_recycling_rewards.py::test_weight_average_never_crashes - ExceptionGroup: Hypothesis found 3 di...
FAILED test_recycling_rewards.py::test_net_points_never_crashes - ExceptionGroup: Hypothesis found 3 distin...
FAILED test_recycling_rewards.py::test_parse_collection_result_is_valid_or_none - ExceptionGroup: Hypothesi...
FAILED test_recycling_rewards.py::test_weight_average_is_nonnegative_integer - ValueError: invalid literal ...
FAILED test_recycling_rewards.py::test_net_points_is_finite_and_nonnegative - ExceptionGroup: Hypothesis fo...
FAILED test_recycling_rewards.py::test_net_points_valid_and_clamped_inputs - AssertionError: assert inf == 0.0
==================================== 7 failed, 2 passed in 0.44s ====================================
```

Step 4 (cont.) - Edge cases discovered

Minimal failing inputs Hypothesis shrank to (extracted from the full pytest traceback output).

PowerShell - Lab 7 - Hypothesis falsifying examples

```
Minimal failing inputs found by Hypothesis (starter code):

test_parse_collection_never_crashes
    input  -> value=b''
    error  -> TypeError: a bytes-like object is required, not 'str'

test_parse_collection_never_crashes
    input  -> value=''
    error  -> ValueError: not enough values to unpack (expected 2, got 1)

test_parse_collection_never_crashes
    input  -> value=None
    error  -> AttributeError: 'NoneType' object has no attribute 'split'

test_weight_average_never_crashes
    input  -> value=b''
    error  -> TypeError: a bytes-like object is required, not 'str'

test_weight_average_never_crashes
    input  -> value=''
    error  -> ValueError: invalid literal for int() with base 10: ''

test_weight_average_never_crashes
    input  -> value=None
    error  -> AttributeError: 'NoneType' object has no attribute 'split'

test_net_points_never_crashes
    input  -> value=b''
    error  -> TypeError: a bytes-like object is required, not 'str'

test_net_points_never_crashes
    input  -> value=''
    error  -> ValueError: not enough values to unpack (expected 2, got 1)

test_net_points_never_crashes
    input  -> value=None
    error  -> AttributeError: 'NoneType' object has no attribute 'split'

test_parse_collection_result_is_valid_or_none
    input  -> value=':'
    error  -> ValueError: could not convert string to float: ''

test_parse_collection_result_is_valid_or_none
    input  -> value=''
    error  -> ValueError: not enough values to unpack (expected 2, got 1)

test_weight_average_is_nonnegative_integer
    input  -> value=''
    error  -> ValueError: invalid literal for int() with base 10: ''

test_net_points_is_finite_and_nonnegative
    input  -> value='|'
    error  -> ValueError: could not convert string to float: ''

test_net_points_is_finite_and_nonnegative
    input  -> value=''
    error  -> ValueError: not enough values to unpack (expected 2, got 1)

test_net_points_valid_and_clamped_inputs
    input  -> '1e308|0'
    error  -> AssertionError: net_points returned inf (overflow)
```

Step 5 - Fixed application run

Fixed recycling_rewards.py: valid input gives the same results; malformed input is handled gracefully (None / 0 / 0.0) instead of crashing.

PowerShell - Lab 7 - fixed recycling_rewards.py

```
> python recycling_rewards.py   (valid inputs)
==== Running parse_collection ====
Enter material:weight in kg (e.g. paper:12): Material and Weight: ('Paper', 12.0)

==== Running weight_average ====
Enter weights in kg as CSV (e.g. 5,12,20): Average Weight: 12

==== Running net_points ====
Enter points|penalty (e.g. 80|10): Net Points: 150.0


> python recycling_rewards.py   (malformed inputs: "paper", "5,,x", "abc")
==== Running parse_collection ====
Enter material:weight in kg (e.g. paper:12): Material and Weight: None

==== Running weight_average ====
Enter weights in kg as CSV (e.g. 5,12,20): Average Weight: 0

==== Running net_points ====
Enter points|penalty (e.g. 80|10): Net Points: 0.0
```

Step 6 - Fuzz tests passing on fixed code

Re-running test_recycling_rewards.py against the fixed code: all 9 tests pass.

PowerShell - Lab 7 - pytest (fixed code)

```
> python -m pytest -v test_recycling_rewards.py
========================================= test session starts =========================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python313\python.exe
hypothesis profile 'default'
rootdir: C:\SEM5\SE\L6_L7_50_recycling_rewards\Lab 7
plugins: hypothesis-6.170.15
collecting ... collected 9 items

test_recycling_rewards.py::test_parse_collection_never_crashes PASSED                    [ 11%]
test_recycling_rewards.py::test_weight_average_never_crashes PASSED                      [ 22%]
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test_recycling_rewards.py::test_net_points_valid_and_clamped_inputs PASSED               [100%]

========================================== 9 passed in 2.05s ==========================================
```